

Unit 1 The Core Principles of Chemistry	21
Unit 2 Application of Core Principles of Chemistry	33
Unit 3 Chemistry Laboratory Skills I	47
Unit 4 General Principles of Chemistry I – Rates, Equilibria and Further Organic Chemistry	51
Unit 5 General Principles of Chemistry II – Transition Metals and Organic Nitrogen Chemistry	65
Unit 6 Chemistry Laboratory Skills II	77

Course structure

- Edexcel's GCE in Chemistry comprises six units and contains an Advanced Subsidiary subset of three AS units.
- The Advanced Subsidiary GCE is the first half of the GCE course and consists of Units 1, 2 and 3. It may be awarded as a discrete qualification or contribute 50 per cent of the total Advanced Level GCE marks.
- The full Advanced Level GCE award consists of the three AS units (Units 1, 2 and 3), plus three A2 units (Units 4, 5 and 6) which make up the other 50 per cent of the Advanced GCE. Students wishing to take the full Advanced Level GCE must, therefore, complete all six units.
- The structure of this qualification allows teachers to construct a course of study which can be taught and assessed either as:
 - ◆ distinct modules of teaching and learning with related units of assessment taken at appropriate stages during the course; or
 - ◆ a linear course which is assessed in its entirety at the end.

1.1 Unit description

Chemical ideas

This unit provides opportunities for students to develop the basic chemical skills of formulae writing, equation writing and calculating chemical quantities.

The study of energetics in chemistry is of theoretical and practical importance. In this unit students learn to define, measure and calculate enthalpy changes. They will see how a study of enthalpy changes can help chemists to understand chemical bonding.

The study of atomic structure introduces s, p, and d orbitals and shows how a more detailed understanding of electron configurations can account for the arrangement of elements in the periodic table.

The unit introduces the three types of strong chemical bonding (ionic, covalent and metallic).

Organic chemistry is also introduced, with students studying alkanes and alkenes.

How chemists work

Practical work measuring energy changes helps students to understand the ideas of uncertainty in measurements and evaluate their results in terms of systematic and random errors.

The study of atomic structure gives some insight into the types of evidence which scientists use to study electrons in atoms. This leads to an appreciation of one of the central features of chemistry which is the explanation of the properties of elements and the patterns in the periodic table in terms of atomic structure.

The role of instrumentation in analytical chemistry is illustrated by mass spectrometry.

Students are introduced to some of the evidence which will help them to understand the different kinds of chemical bonding.

Chemists set up theoretical models and gain insights by comparing real and ideal properties of chemicals. This is illustrated in the unit by the ionic model and the comparison of lattice energies calculated from theory with those determined with the help of Born-Haber cycles.

Throughout the unit students see the importance of chemical data and learn to select data from databases and use it to look for patterns and calculate other quantities.

The introduction to organic chemistry shows how chemists work safely with potentially hazardous chemicals by managing risks.

Chemistry in action

The uses of mass spectrometry illustrate the importance of sensitive methods of analysis in areas such as space research, medical research and diagnosis, in detecting drugs in sport and in environmental monitoring.

In this unit students learn how chemical insights can help to make the use of polymeric and other materials more sustainable. This involves analysis of the uses of energy, raw materials and other resources at each stage of the life cycle of products.